

System Driven Hardware Design

- System, Amplifier, PCB – Development and Sequencing -

International Master of Science
SS23

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Agenda

- o Course Outline
- o Marking
- o Problem Statement
- o System Overview
- o Short Introduction of the Hardware and Amplifier Schematic
- o Sequencing

Course Outline

Version 1.0 /28.03.23 - Bw

2023 - week	15	16	17	18	19	20	21	22	23	24	25	26	27	28	28
lecture week	1	2	3	4	5	6	7	8	9		11	12	13	14	
Lecture Content SS23	Course Organisation, System, Amplifier, Sequencing	Doppler-Shift BP 4th Order Derivation	24 GHz Radar	PSoC Heart Beat / 3-Bit Counter	ADC FS Range and PSoC UART	DFT	DFT	Windowing	Windowing	CFAR Algorithmn	CFAR Algorithmn	Fixed Point	Testing	Puffer	
Comments SS22	erster Foliensatz perfekt.	Starpoint Konzept, SMD LEDs, Doppler-Shift: Wdh. Für die Herleitung keine Zeit.	Hinweis: Breadboard aufbau starten												
SWS	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
Production				Tape In (Friday)	Tape In (Friday)	Manfuactoring	PCB back on Friday								
Development Phases															
Lab Content SS23		Layout		PCB Review / PSoC Q&A		State Machine, ADC and UART Software testing		DFT Software Integration and Soldering		CFAR Algorithm Integration and Start Up		Presentations			
Monday (Ax/By)		A	B	Bankholiday	A	B	A		B	A	B	A	B	A	B
Wednesday (Cx/Dy)		D	C	D	C	D	C	D	C	D	E	D	E		E
Thursday (Ex/Fy/Gy)		F	E	F	E	Bankholiday	F	E	Bankholiday	F	C	F	C	F	
		G		G		G			Bankholiday	G		G		G	
2023-week	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29

Appointment	Topic	Upload <u>before</u> the lab	Outcome
1	Layout	Finished schematic entry and footprints are placed	Finished layout
2	PCB Review / PSoC Q&A	Finished layout, BOM, PCB Specification (Elekonta).	Gerber Files for Elekonta
3	Testing of: State Machine, ADC and UART	Software: Statemachine, ADC sampling, UART Transfer. Hardware: Bring completely wired Bread Board	- Transferring Amplifier Output Signal to Matlab - Testing 1. Filter Stage
4	DFT Software Integration and Soldering	Software code including DFT in C or Matlab	- Assemble your PCBs and Start-Up - DFT ist working
5	CFAR Algorithm Integration and Start Up	upload reviewed software code and finished soldered board	- Add the 24GHz Radar Sensor to your PCB and FreeSoc - Present your overall results
6	Presentation		

- You can achieve 100 points in the module
- Contribution to the overall module mark
 - Lab counts 40 points
 - Exam counts 60 points
- The Lab and the exam has to be passed independently.
This means you have to achieve 40% in **Lab (16 points)** and 40% in **Exam (24 points)**.

- Duration: 90 Minutes
- 4 Questions with 15 points each for 20 Minutes each
- 10 Minutes Reading Time
- Possible topics covered in the questions
 - PCB
 - Software
 - Signal Processing
 - System Considerations

Points from $\geq$	Points until $<$	Mark
-2	-1	0
0,00	40,00	5,0
40,00	42,00	4,0
42,00	47,00	3,7
47,00	54,00	3,3
54,00	59,00	3,0
59,00	64,00	2,7
64,00	71,00	2,3
71,00	76,00	2,0
76,00	81,00	1,7
81,00	87,00	1,3
87,00	100,00	1,0

Problem Statement: We want to measure speed of a human.

A) Possible Measurement technique: (**physical principal** / technical solution)

- Answer the question: Can we identify a **key component**?
(i.e. without it the system cannot be developed, here radar module)
- Which technology?
(Electromagnetic waves with constant frequency CW)
- Which Frequency range?
(24 GHz / 12GHz / 79GHz **legal aspects** ISM bands)

B) Market Research:

- 1. possibility 24 GHz CW-Radar Module (output signal IQ/SSB/Baseband-> further signal processing, complexity / reliability / costs)
- 2. possibility: 74GHz

C) Develop concepts for all possible solutions

- Signal Flow Diagram

- System Aspects: Expected Frequency Range to be processed? (**complexity**)
- This complexity determines the further need HW and SW parts. (**costs / complexity**)

Underlying
Principal

Different
Concepts

Evaluate

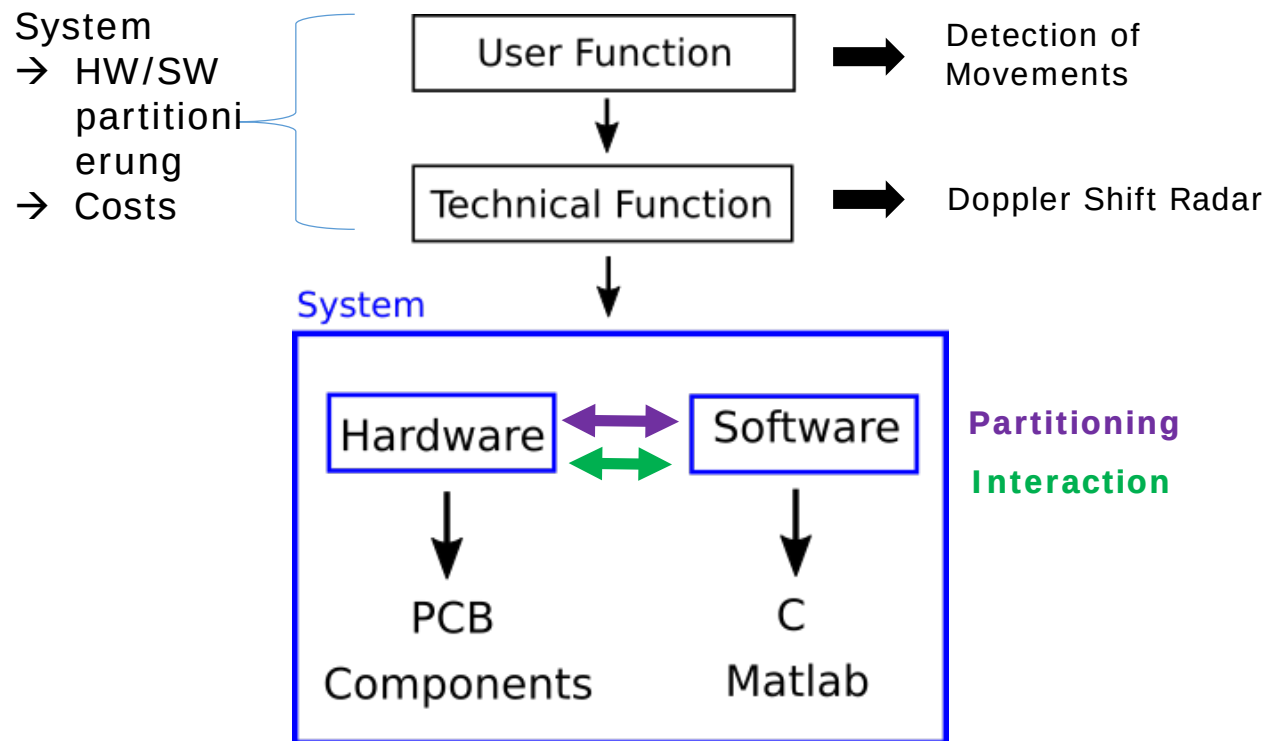
Constraints
beyond the
underlying
principal

Iterations based on:

1. Legal aspects
2. Technical
realizable
3. Development
complexity
4. Costs

System function for the user: A 24 GHz radar detects movements and signals this by turning on a red LED, as long as a movement is detected.

Lab Realisation



Your Task

Used Hardware

- 24 GHz CW-Radar sensor
- Operational Amplifier
- FreeSoc 2 (PScC5LP)

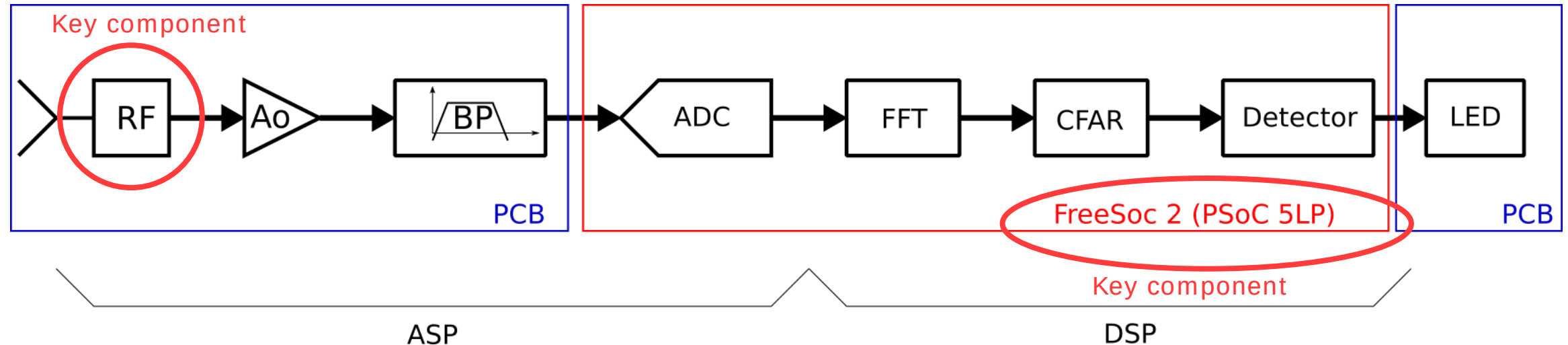
Used Software

- Matlab
- KiCad
- PSoC Creator
- LT-Spice

Aim

- Build Up a System Concept,
- Break Down the Requirements
- Circuit, PCB and Software Development

System function for the user: A 24 GHz radar detects movements and signals this by turning on a red LED, as long as a movement is detected.

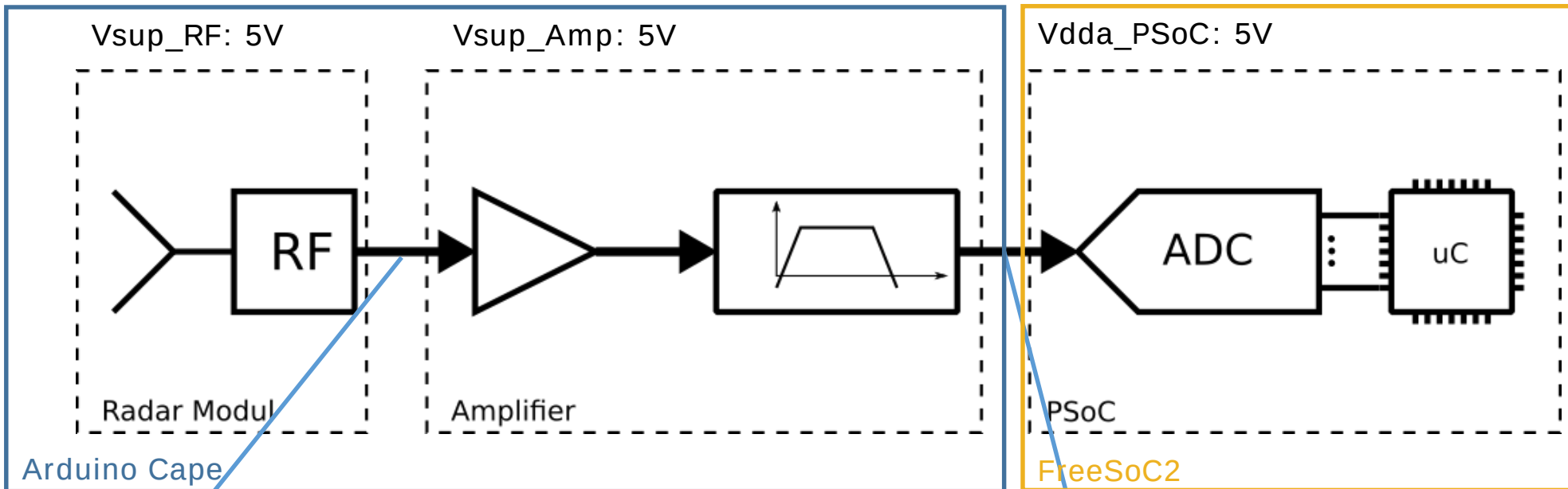


RF: Radar Modul
Ao: Amplification
BP: Bandpassfilter
ADC: Analog to Digital Converter
FFT: Fast Fourier Transform
CFAR: Constant False Alarm Rate
Detector: Detects and Tracks Movements
LED: HMI shows a Movement

Task
Draw an initial Signal Flow Diagram
Assume First Systemparameter

Question

1. Which Interfaces do we have?
2. Which systemparameters do we need?



-Pegelplan
-Initial BOM

V_LF

Voltage level	Min. 300uV / max. ? – single ended
Frequency	< 100 MHz
Coupling	AC

-Pv

Amplifier:
Passband:
Amplification:
AC-Coupling:
vdda/2
Supply Voltage: 5V

4th order BP
60Hz – 800Hz
?
mean voltage is

V_ADC

Input Voltage range	100mV – 4.75V, single ended
Frequency	60Hz - 800 Hz
Coupling	DC

ADC

Nyquist rate
1.6 kHz

Description

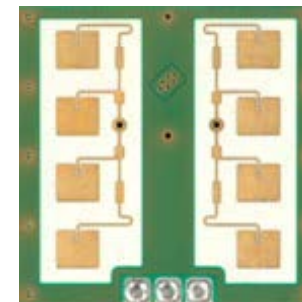
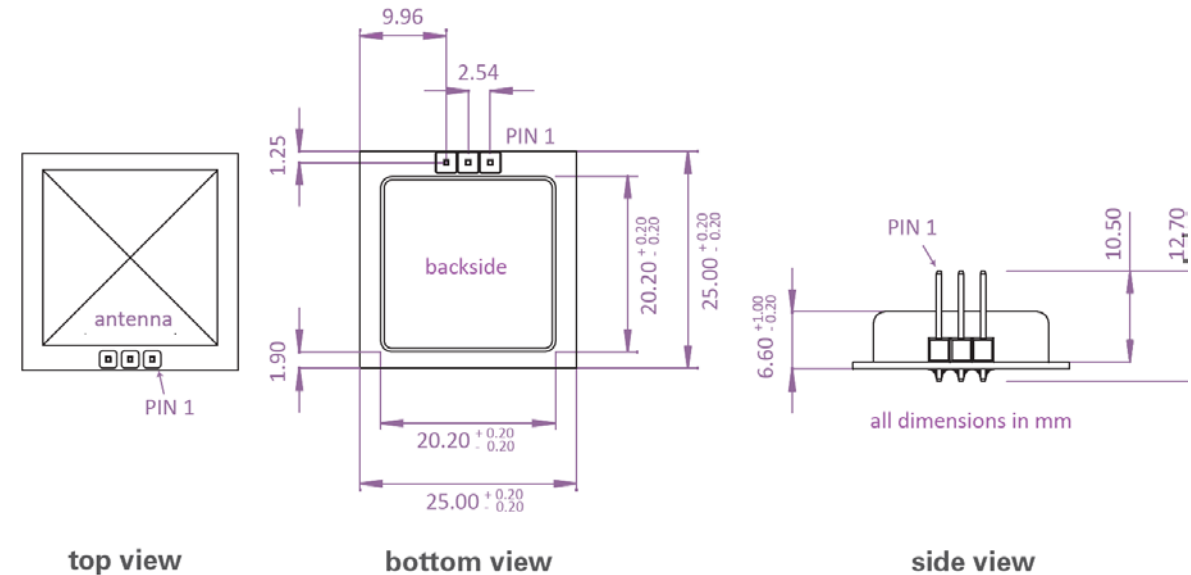
The IPM-165 is a 24GHz Doppler module with an asymmetrical wide beam for detection of moving objects.

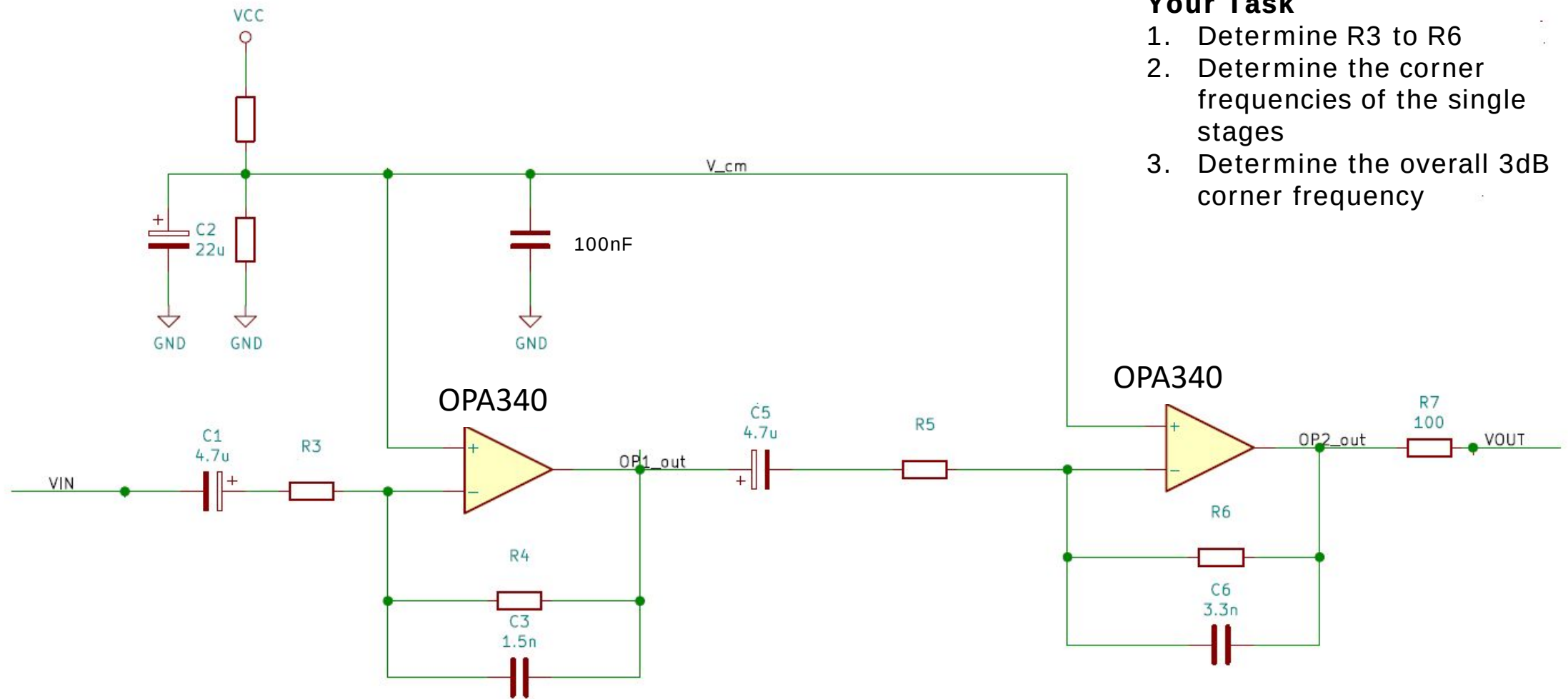
Interface

The sensor provides a 2.54mm grid, single row pin header (square pin $\square 0.635\text{mm}$)

PIN #	DESCRIPTION	IN / OUT	COMMENT
1	V_{CC}	input	supply voltage (4.75 - 5.25V)
2	IF1	output	signal I(nphase)
3	GND	input	analog ground

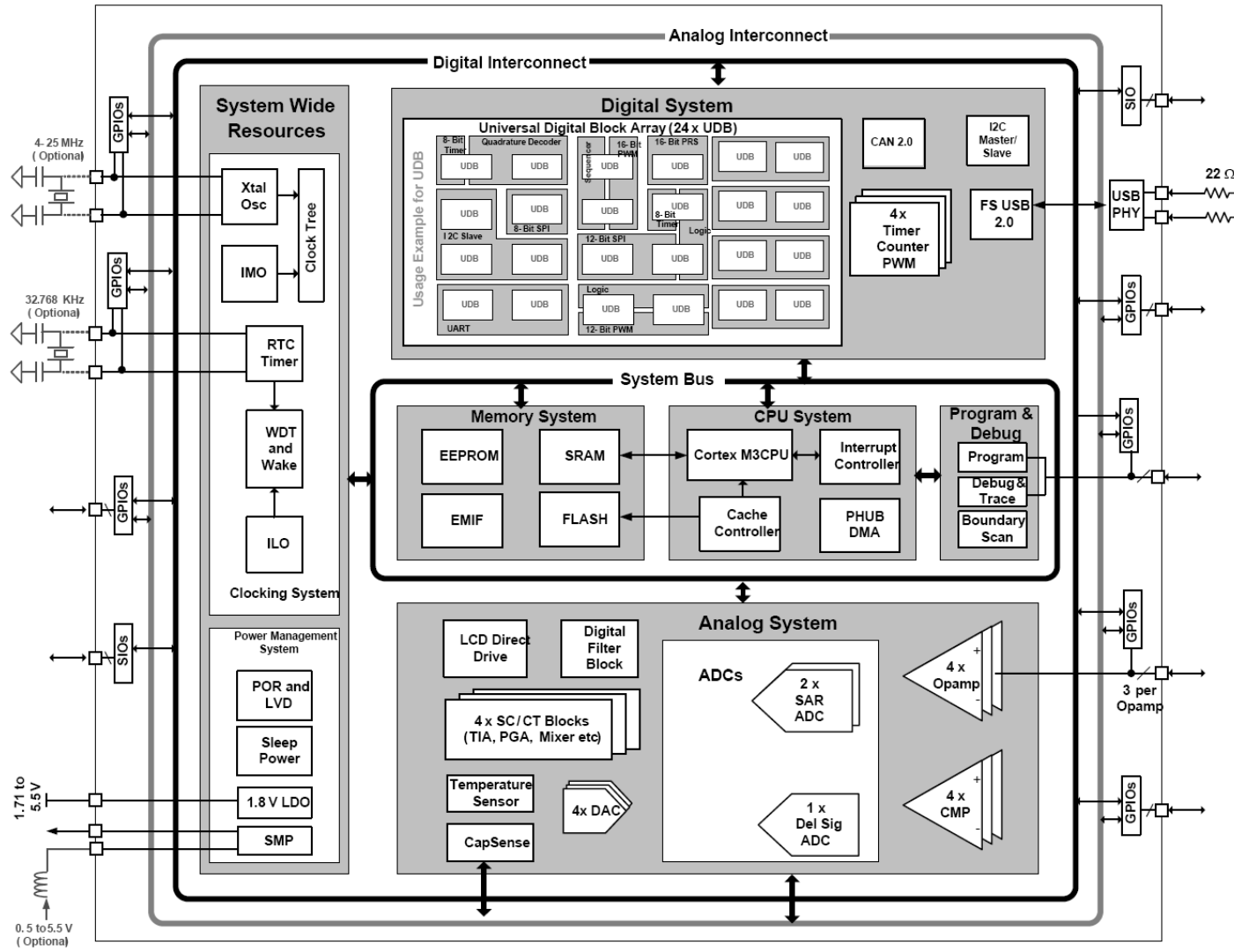
IPM 165 Mechanical Outlines





Your Task

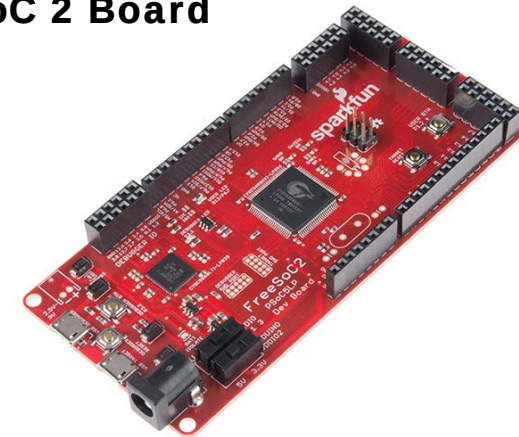
1. Determine R3 to R6
2. Determine the corner frequencies of the single stages
3. Determine the overall 3dB corner frequency

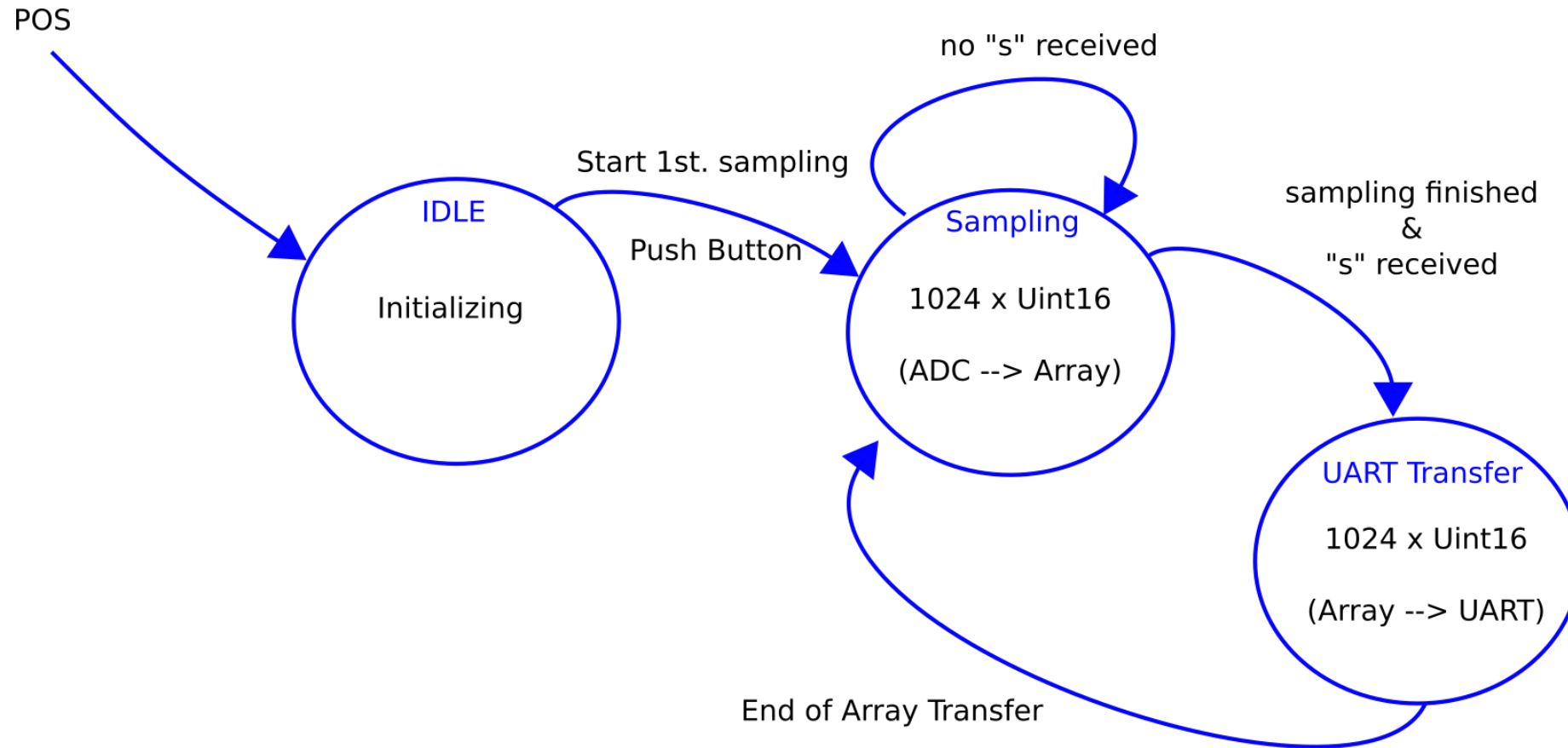


PSoC 5LP

- 32-bit ARM Cortex-M3 core plus DMA controller and digital filter processor, at up to 80 MHz
- Analog peripherals
- Configurable 8- to 20-bit delta-sigma ADC
- Up to two 12-bit SAR ADCs
- Four 8-bit DACs
- Four comparators
- Four opamps
- Four programmable analog blocks, to create:
 - Programmable gain amplifier (PGA)
 - Transimpedance amplifier (TIA)
 - Mixer

FreeSoC 2 Board





Use a **switch-case** structure to implement the state machine